THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 7 & 8: **How do Organisms Reproduce?, Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Fertilisation results in the formation of: 1

(a) spore **(b)** gamete **(c)** zygote **(d)** bud

2. A student states: “Oral contraceptive pills change the hormonal balance to prevent eggs being released.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only with copper-T **(d)** Cannot be determined

3. The plant in which reproduction by fragmentation is common is: 1

(a) rose **(b)** Spirogyra **(c)** potato **(d)** mango

4. The fluid that carries sperms is formed by secretions from: 1

(a) ovaries **(b)** seminal vesicles and prostate

(c) liver **(d)** kidneys

5. For DNA copying, Student A says it produces exactly identical copies; Student B says it produces copies with slight variations. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The Y chromosome is found in: 1

(a) males only **(b)** females only **(c)** both **(d)** neither

7. A student states: “A cross of TT × Tt gives some short plants.” The correct response is: 1

(a) Correct **(b)** Incorrect — all are tall

(c) Correct, 25% short **(d)** Cannot be determined

8. Mendel studied how many pairs of contrasting characters in pea? 1

(a) nine **(b)** five **(c)** seven **(d)** two

9. The chromosome number is restored at: 1

(a) gamete formation **(b)** growth **(c)** fertilisation **(d)** germination

10. For the outcome of Tt × tt, Student A says 1 tall : 1 short; Student B says all tall. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What are the functions of the ovary in human females? 2

12. Why does menstruation occur? 2

13. Why do offspring resemble their parents but are not identical? 2

14. What is meant by 'pure line'? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe multiple fission in Plasmodium. 3

16. How many types of gametes can a plant with genotype TtRr produce? List them. 3

17. What are the agents of pollination? Give one example of each. 3

OR

How is the number of chromosomes maintained across generations in sexual reproduction? 3

18. In humans, what fraction of children of a couple will be boys? Explain. 3

OR

What is meant by DNA? Where is it found and what is its role? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A pregnant woman's diet affects the growth of the foetus. 5

(i) Through what does the foetus get nutrition?
(ii) What are villi?
(iii) What substances pass from mother to embryo?
(iv) What passes from embryo to mother?
(v) How long is the development period?

OR

(i) What are the different contraceptive methods? 5
(ii) Explain how each works.
(iii) Why should contraceptive methods be adopted?

20. **Case study:** Mendel crossed a violet-flowered pea plant with a white-flowered one. All F1 had violet flowers. 5

(i) Which trait is dominant?
(ii) What will be the F2 ratio?
(iii) Write the genotypes of violet plants in F2.
(iv) How can you find if a violet plant is pure?
(v) What is the white flower genotype?

OR

(i) Explain the dihybrid cross with a Punnett square. 5
(ii) What are parental and new (recombinant) combinations?

END OF QUESTION PAPER — Total: 40 Marks